

Import Library

```
In [12]: import pandas as pd
```

Import Data

```
In [13]: data = pd.read_csv (r"D:\Downloads\Udemy-PY\Project+1+--+Weather+Dataset.csv")
```

Cheking the Data

```
In [14]: data
```

Out[14]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Wea
0	1/1/2012 0:00	-1.8	-3.9	86	4	8.0	101.24	
1	1/1/2012 1:00	-1.8	-3.7	87	4	8.0	101.24	
2	1/1/2012 2:00	-1.8	-3.4	89	7	4.0	101.26	Free Drizzle
3	1/1/2012 3:00	-1.5	-3.2	88	6	4.0	101.27	Free Drizzle
4	1/1/2012 4:00	-1.5	-3.3	88	7	4.8	101.23	
...	...	...	...	...	...	...	...	
8779	12/31/2012 19:00	0.1	-2.7	81	30	9.7	100.13	S
8780	12/31/2012 20:00	0.2	-2.4	83	24	9.7	100.03	S
8781	12/31/2012 21:00	-0.5	-1.5	93	28	4.8	99.95	S
8782	12/31/2012 22:00	-0.2	-1.8	89	28	9.7	99.91	S
8783	12/31/2012 23:00	0.0	-2.1	86	30	11.3	99.89	S

8784 rows × 8 columns



Checking Data Frames

```
In [15]: data.head()
```

Out[15]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weather
0	1/1/2012 0:00	-1.8	-3.9	86	4	8.0	101.24	Fog
1	1/1/2012 1:00	-1.8	-3.7	87	4	8.0	101.24	Fog
2	1/1/2012 2:00	-1.8	-3.4	89	7	4.0	101.26	Freezing Drizzle,Fog
3	1/1/2012 3:00	-1.5	-3.2	88	6	4.0	101.27	Freezing Drizzle,Fog
4	1/1/2012 4:00	-1.5	-3.3	88	7	4.8	101.23	Fog

```
In [16]: data.shape
```

Out[16]: (8784, 8)

```
In [17]: data.index
```

Out[17]: RangeIndex(start=0, stop=8784, step=1)

```
In [18]: data.columns
```

Out[18]: Index(['Date/Time', 'Temp_C', 'Dew Point Temp_C', 'Rel Hum_%', 'Wind Speed_km/h', 'Visibility_km', 'Press_kPa', 'Weather'], dtype='object')

```
In [19]: data.dtypes
```

Out[19]: Date/Time object
Temp_C float64
Dew Point Temp_C float64
Rel Hum_% int64
Wind Speed_km/h int64
Visibility_km float64
Press_kPa float64
Weather object
dtype: object

```
In [20]: data['Weather'].unique ()
```

```
Out[20]: array(['Fog', 'Freezing Drizzle,Fog', 'Mostly Cloudy', 'Cloudy', 'Rain',
               'Rain Showers', 'Mainly Clear', 'Snow Showers', 'Snow', 'Clear',
               'Freezing Rain,Fog', 'Freezing Rain', 'Freezing Drizzle',
               'Rain,Snow', 'Moderate Snow', 'Freezing Drizzle,Snow',
               'Freezing Rain,Snow Grains', 'Snow,Blowing Snow', 'Freezing Fog',
               'Haze', 'Rain,Fog', 'Drizzle,Fog', 'Drizzle',
               'Freezing Drizzle,Haze', 'Freezing Rain,Haze', 'Snow,Haze',
               'Snow,Fog', 'Snow,Ice Pellets', 'Rain,Haze', 'Thunderstorms,Rain',
               'Thunderstorms,Rain Showers', 'Thunderstorms,Heavy Rain Showers',
               'Thunderstorms,Rain Showers,Fog', 'Thunderstorms',
               'Thunderstorms,Rain,Fog',
               'Thunderstorms,Moderate Rain Showers,Fog', 'Rain Showers,Fog',
               'Rain Showers,Snow Showers', 'Snow Pellets', 'Rain,Snow,Fog',
               'Moderate Rain,Fog', 'Freezing Rain,Ice Pellets,Fog',
               'Drizzle,Ice Pellets,Fog', 'Drizzle,Snow', 'Rain,Ice Pellets',
               'Drizzle,Snow,Fog', 'Rain,Snow Grains', 'Rain,Snow,Ice Pellets',
               'Snow Showers,Fog', 'Moderate Snow,Blowing Snow'], dtype=object)
```

```
In [22]: data.nunique()
```

```
Out[22]: Date/Time      8784
         Temp_C         533
         Dew Point Temp_C 489
         Rel Hum_%       83
         Wind Speed_km/h  34
         Visibility_km    24
         Press_kPa       518
         Weather         50
         dtype: int64
```

```
In [23]: data.count()
```

```
Out[23]: Date/Time      8784
         Temp_C         8784
         Dew Point Temp_C 8784
         Rel Hum_%       8784
         Wind Speed_km/h  8784
         Visibility_km    8784
         Press_kPa       8784
         Weather         8784
         dtype: int64
```

```
In [25]: data['Weather'].value_counts()
```

```

Out[25]: Weather
Mainly Clear                2106
Mostly Cloudy              2069
Cloudy                     1728
Clear                      1326
Snow                       390
Rain                       306
Rain Showers               188
Fog                        150
Rain,Fog                   116
Drizzle,Fog                80
Snow Showers               60
Drizzle                    41
Snow,Fog                   37
Snow,Blowing Snow         19
Rain,Snow                  18
Thunderstorms,Rain Showers 16
Haze                       16
Drizzle,Snow,Fog          15
Freezing Rain              14
Freezing Drizzle,Snow      11
Freezing Drizzle           7
Freezing Drizzle,Fog       6
Snow,Ice Pellets           6
Snow,Haze                  5
Freezing Rain,Fog          4
Moderate Snow              4
Rain,Snow,Ice Pellets      4
Freezing Fog               4
Snow Showers,Fog           4
Thunderstorms,Rain Showers,Fog 3
Thunderstorms,Rain         3
Freezing Drizzle,Haze      3
Rain,Haze                  3
Drizzle,Snow               2
Moderate Snow,Blowing Snow 2
Thunderstorms              2
Freezing Rain,Haze         2
Rain Showers,Snow Showers  2
Freezing Rain,Snow Grains  1
Thunderstorms,Heavy Rain Showers 1
Thunderstorms,Moderate Rain Showers,Fog 1
Thunderstorms,Rain,Fog     1
Freezing Rain,Ice Pellets,Fog 1
Moderate Rain,Fog          1
Rain,Snow,Fog              1
Snow Pellets               1
Rain Showers,Fog           1
Drizzle,Ice Pellets,Fog    1
Rain,Ice Pellets           1
Rain,Snow Grains           1
Name: count, dtype: int64

```

```
In [26]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8784 entries, 0 to 8783
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Date/Time              8784 non-null   object
1   Temp_C                 8784 non-null   float64
2   Dew Point Temp_C       8784 non-null   float64
3   Rel Hum_%              8784 non-null   int64
4   Wind Speed_km/h        8784 non-null   int64
5   Visibility_km           8784 non-null   float64
6   Press_kPa              8784 non-null   float64
7   Weather                8784 non-null   object
dtypes: float64(4), int64(2), object(2)
memory usage: 549.1+ KB
```

Starting answers to the Project

1. Find all unique 'Wind Speed' values

In [27]: `data.head(2)`

Out[27]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weather
0	1/1/2012 0:00	-1.8	-3.9	86	4	8.0	101.24	Fog
1	1/1/2012 1:00	-1.8	-3.7	87	4	8.0	101.24	Fog

In [28]: `data.nunique()`

Out[28]:

Date/Time	8784
Temp_C	533
Dew Point Temp_C	489
Rel Hum_%	83
Wind Speed_km/h	34
Visibility_km	24
Press_kPa	518
Weather	50

dtype: int64

In [41]: `data['Wind Speed_km/h'].unique() #Answer`

Out[41]: array([4, 7, 6, 9, 15, 13, 20, 22, 19, 24, 30, 35, 39, 32, 33, 26, 44, 43, 48, 37, 28, 17, 11, 0, 83, 70, 57, 46, 41, 52, 50, 63, 54, 2])

The info bellow is Not Required

In [34]: `data['Wind Speed_km/h'].max()`

Out[34]: np.int64(83)

In [38]: data['Wind Speed_km/h'].std()

Out[38]: np.float64(8.688696079468851)

In [39]: data['Wind Speed_km/h'].mean()

Out[39]: np.float64(14.94546903460838)

2. Find exact numers of times when weather is 'Exactly Clear'

In [42]: data.head(2)

Out[42]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weather
0	1/1/2012 0:00	-1.8	-3.9	86	4	8.0	101.24	Fog
1	1/1/2012 1:00	-1.8	-3.7	87	4	8.0	101.24	Fog

In [46]: *#by Value_Counts funtion*
data['Weather'].value_counts()

```
Out[46]: Weather
Mainly Clear                2106
Mostly Cloudy              2069
Cloudy                     1728
Clear                      1326
Snow                       390
Rain                       306
Rain Showers               188
Fog                        150
Rain,Fog                   116
Drizzle,Fog                80
Snow Showers               60
Drizzle                    41
Snow,Fog                   37
Snow,Blowing Snow         19
Rain,Snow                  18
Thunderstorms,Rain Showers 16
Haze                       16
Drizzle,Snow,Fog          15
Freezing Rain              14
Freezing Drizzle,Snow      11
Freezing Drizzle           7
Freezing Drizzle,Fog       6
Snow,Ice Pellets           6
Snow,Haze                  5
Freezing Rain,Fog          4
Moderate Snow              4
Rain,Snow,Ice Pellets      4
Freezing Fog               4
Snow Showers,Fog           4
Thunderstorms,Rain Showers,Fog 3
Thunderstorms,Rain         3
Freezing Drizzle,Haze      3
Rain,Haze                  3
Drizzle,Snow               2
Moderate Snow,Blowing Snow 2
Thunderstorms              2
Freezing Rain,Haze         2
Rain Showers,Snow Showers  2
Freezing Rain,Snow Grains  1
Thunderstorms,Heavy Rain Showers 1
Thunderstorms,Moderate Rain Showers,Fog 1
Thunderstorms,Rain,Fog     1
Freezing Rain,Ice Pellets,Fog 1
Moderate Rain,Fog          1
Rain,Snow,Fog              1
Snow Pellets               1
Rain Showers,Fog           1
Drizzle,Ice Pellets,Fog    1
Rain,Ice Pellets           1
Rain,Snow Grains           1
Name: count, dtype: int64
```

```
In [49]: #by Filtering method
data[data.Weather == 'Clear']
```

Out[49]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weath
67	1/3/2012 19:00	-16.9	-24.8	50	24	25.0	101.74	Cle
114	1/5/2012 18:00	-7.1	-14.4	56	11	25.0	100.71	Cle
115	1/5/2012 19:00	-9.2	-15.4	61	7	25.0	100.80	Cle
116	1/5/2012 20:00	-9.8	-15.7	62	9	25.0	100.83	Cle
117	1/5/2012 21:00	-9.0	-14.8	63	13	25.0	100.83	Cle
...	...	...	...	...	...	...	...	...
8646	12/26/2012 6:00	-13.4	-14.8	89	4	25.0	102.47	Cle
8698	12/28/2012 10:00	-6.1	-8.6	82	19	24.1	101.27	Cle
8713	12/29/2012 1:00	-11.9	-13.6	87	11	25.0	101.31	Cle
8714	12/29/2012 2:00	-11.8	-13.1	90	13	25.0	101.33	Cle
8756	12/30/2012 20:00	-13.8	-16.5	80	24	25.0	101.52	Cle

1326 rows × 8 columns



```
In [50]: data.groupby('Weather').get_group('Clear')
```


Out[50]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weath
67	1/3/2012 19:00	-16.9	-24.8	50	24	25.0	101.74	Cle
114	1/5/2012 18:00	-7.1	-14.4	56	11	25.0	100.71	Cle
115	1/5/2012 19:00	-9.2	-15.4	61	7	25.0	100.80	Cle
116	1/5/2012 20:00	-9.8	-15.7	62	9	25.0	100.83	Cle
117	1/5/2012 21:00	-9.0	-14.8	63	13	25.0	100.83	Cle
...	...	...	...	...	...	...	...	...
8646	12/26/2012 6:00	-13.4	-14.8	89	4	25.0	102.47	Cle
8698	12/28/2012 10:00	-6.1	-8.6	82	19	24.1	101.27	Cle
8713	12/29/2012 1:00	-11.9	-13.6	87	11	25.0	101.31	Cle
8714	12/29/2012 2:00	-11.8	-13.1	90	13	25.0	101.33	Cle
8756	12/30/2012 20:00	-13.8	-16.5	80	24	25.0	101.52	Cle

1326 rows × 8 columns



3. Find the number of rows where 'Wind Speed = 4km/h'

```
In [54]: data[data['Wind Speed_km/h'] == 4]
```

Out[54]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weatl
0	1/1/2012 0:00	-1.8	-3.9	86	4	8.0	101.24	F
1	1/1/2012 1:00	-1.8	-3.7	87	4	8.0	101.24	F
96	1/5/2012 0:00	-8.8	-11.7	79	4	9.7	100.32	Sn
101	1/5/2012 5:00	-7.0	-9.5	82	4	4.0	100.19	Sn
146	1/7/2012 2:00	-8.1	-11.1	79	4	19.3	100.15	Clo
...	...	...	...	...	...	...	...	
8768	12/31/2012 8:00	-8.6	-10.3	87	4	3.2	101.14	Sn Show
8769	12/31/2012 9:00	-8.1	-9.6	89	4	2.4	101.09	Sn
8770	12/31/2012 10:00	-7.4	-8.9	89	4	6.4	101.05	Snow,F
8772	12/31/2012 12:00	-5.8	-7.5	88	4	12.9	100.78	Sn
8773	12/31/2012 13:00	-4.6	-6.6	86	4	12.9	100.63	Sn

474 rows × 8 columns



4. Find all Null values, If Any!

```
In [56]: data.isnull().sum()
```

```
Out[56]: Date/Time      0
Temp_C              0
Dew Point Temp_C    0
Rel Hum_%           0
Wind Speed_km/h     0
Visibility_km        0
Press_kPa           0
Weather             0
dtype: int64
```

```
In [57]: data.notnull().sum()
```

```
Out[57]: Date/Time      8784
Temp_C      8784
Dew Point Temp_C  8784
Rel Hum_%    8784
Wind Speed_km/h 8784
Visibility_km 8784
Press_kPa    8784
Weather      8784
dtype: int64
```

5. Rename 'Weather' to 'Weather Condition'

```
In [59]: data.rename(columns = {'Weather':'Weather Condition'}) #Temporarily
```

Out[59]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weather Condition
0	1/1/2012 0:00	-1.8	-3.9	86	4	8.0	101.24	
1	1/1/2012 1:00	-1.8	-3.7	87	4	8.0	101.24	
2	1/1/2012 2:00	-1.8	-3.4	89	7	4.0	101.26	Free Drizzle
3	1/1/2012 3:00	-1.5	-3.2	88	6	4.0	101.27	Free Drizzle
4	1/1/2012 4:00	-1.5	-3.3	88	7	4.8	101.23	
...	...	...	...	...	...	...	...	
8779	12/31/2012 19:00	0.1	-2.7	81	30	9.7	100.13	S
8780	12/31/2012 20:00	0.2	-2.4	83	24	9.7	100.03	S
8781	12/31/2012 21:00	-0.5	-1.5	93	28	4.8	99.95	S
8782	12/31/2012 22:00	-0.2	-1.8	89	28	9.7	99.91	S
8783	12/31/2012 23:00	0.0	-2.1	86	30	11.3	99.89	S

8784 rows × 8 columns



```
In [62]: data.rename(columns = {'Weather':'Weather Condition'}, inplace = True) #Permanently
data.head(2)
```

Out[62]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weather Condition
0	1/1/2012 0:00	-1.8	-3.9	86	4	8.0	101.24	Fog
1	1/1/2012 1:00	-1.8	-3.7	87	4	8.0	101.24	Fog



6. What is the Mean 'Visibility Range'?

In [63]: `data['Visibility_km'].mean()`Out[63]: `np.float64(27.664446721311474)`

7. What is Standard Deviation of 'Pressure' column?

In [65]: `data['Press_kPa'].std()`Out[65]: `np.float64(0.8440047459486483)`

8. What is Variance of 'Relative Humidity'?

In [66]: `data['Rel Hum_%'].var()`Out[66]: `np.float64(286.24855019850196)`

8. Find all instances when 'Snow' was recorded

In [68]: `#Value_Counts
data['Weather Condition'].value_counts()`

```
Out[68]: Weather Condition
Mainly Clear 2106
Mostly Cloudy 2069
Cloudy 1728
Clear 1326
Snow 390
Rain 306
Rain Showers 188
Fog 150
Rain,Fog 116
Drizzle,Fog 80
Snow Showers 60
Drizzle 41
Snow,Fog 37
Snow,Blowing Snow 19
Rain,Snow 18
Thunderstorms,Rain Showers 16
Haze 16
Drizzle,Snow,Fog 15
Freezing Rain 14
Freezing Drizzle,Snow 11
Freezing Drizzle 7
Freezing Drizzle,Fog 6
Snow,Ice Pellets 6
Snow,Haze 5
Freezing Rain,Fog 4
Moderate Snow 4
Rain,Snow,Ice Pellets 4
Freezing Fog 4
Snow Showers,Fog 4
Thunderstorms,Rain Showers,Fog 3
Thunderstorms,Rain 3
Freezing Drizzle,Haze 3
Rain,Haze 3
Drizzle,Snow 2
Moderate Snow,Blowing Snow 2
Thunderstorms 2
Freezing Rain,Haze 2
Rain Showers,Snow Showers 2
Freezing Rain,Snow Grains 1
Thunderstorms,Heavy Rain Showers 1
Thunderstorms,Moderate Rain Showers,Fog 1
Thunderstorms,Rain,Fog 1
Freezing Rain,Ice Pellets,Fog 1
Moderate Rain,Fog 1
Rain,Snow,Fog 1
Snow Pellets 1
Rain Showers,Fog 1
Drizzle,Ice Pellets,Fog 1
Rain,Ice Pellets 1
Rain,Snow Grains 1
Name: count, dtype: int64
```

```
In [70]: #Filtering
data[data['Weather Condition'] == 'Snow']
```

Out[70]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weat Condit
55	1/3/2012 7:00	-14.0	-19.5	63	19	25.0	100.95	Sr
84	1/4/2012 12:00	-13.7	-21.7	51	11	24.1	101.25	Sr
86	1/4/2012 14:00	-11.3	-19.0	53	7	19.3	100.97	Sr
87	1/4/2012 15:00	-10.2	-16.3	61	11	9.7	100.89	Sr
88	1/4/2012 16:00	-9.4	-15.5	61	13	19.3	100.79	Sr
...	...	...	...	...	...	...	...	
8779	12/31/2012 19:00	0.1	-2.7	81	30	9.7	100.13	Sr
8780	12/31/2012 20:00	0.2	-2.4	83	24	9.7	100.03	Sr
8781	12/31/2012 21:00	-0.5	-1.5	93	28	4.8	99.95	Sr
8782	12/31/2012 22:00	-0.2	-1.8	89	28	9.7	99.91	Sr
8783	12/31/2012 23:00	0.0	-2.1	86	30	11.3	99.89	Sr

390 rows × 8 columns



```
In [71]: #GroupBy
data.groupby('Weather Condition').get_group('Snow')
```

Out[71]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weat Condit
55	1/3/2012 7:00	-14.0	-19.5	63	19	25.0	100.95	Sr
84	1/4/2012 12:00	-13.7	-21.7	51	11	24.1	101.25	Sr
86	1/4/2012 14:00	-11.3	-19.0	53	7	19.3	100.97	Sr
87	1/4/2012 15:00	-10.2	-16.3	61	11	9.7	100.89	Sr
88	1/4/2012 16:00	-9.4	-15.5	61	13	19.3	100.79	Sr
...	...	...	...	...	...	...	...	
8779	12/31/2012 19:00	0.1	-2.7	81	30	9.7	100.13	Sr
8780	12/31/2012 20:00	0.2	-2.4	83	24	9.7	100.03	Sr
8781	12/31/2012 21:00	-0.5	-1.5	93	28	4.8	99.95	Sr
8782	12/31/2012 22:00	-0.2	-1.8	89	28	9.7	99.91	Sr
8783	12/31/2012 23:00	0.0	-2.1	86	30	11.3	99.89	Sr

390 rows × 8 columns



```
In [72]: #To consider all values containig string: Snow, we use str.contains
data[data['Weather Condition'].str.contains('Snow')]
```

Out[72]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weat Condit
41	1/2/2012 17:00	-2.1	-9.5	57	22	25.0	99.66	Sr Shov
44	1/2/2012 20:00	-5.6	-13.4	54	24	25.0	100.07	Sr Shov
45	1/2/2012 21:00	-5.8	-12.8	58	26	25.0	100.15	Sr Shov
47	1/2/2012 23:00	-7.4	-14.1	59	17	19.3	100.27	Sr Shov
48	1/3/2012 0:00	-9.0	-16.0	57	28	25.0	100.35	Sr Shov
...	...	...	...	...	...	...	...	
8779	12/31/2012 19:00	0.1	-2.7	81	30	9.7	100.13	Sr
8780	12/31/2012 20:00	0.2	-2.4	83	24	9.7	100.03	Sr
8781	12/31/2012 21:00	-0.5	-1.5	93	28	4.8	99.95	Sr
8782	12/31/2012 22:00	-0.2	-1.8	89	28	9.7	99.91	Sr
8783	12/31/2012 23:00	0.0	-2.1	86	30	11.3	99.89	Sr

583 rows × 8 columns



9. Find all where 'Wind Speed > 24' and 'Visibility = 25'

```
In [73]: data[(data['Wind Speed_km/h'] > 24) & (data['Visibility_km'] == 25)]
```


Out[73]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weat Condit
23	1/1/2012 23:00	5.3	2.0	79	30	25.0	99.31	Clo
24	1/2/2012 0:00	5.2	1.5	77	35	25.0	99.26	F Shov
25	1/2/2012 1:00	4.6	0.0	72	39	25.0	99.26	Clo
26	1/2/2012 2:00	3.9	-0.9	71	32	25.0	99.26	Mc Clo
27	1/2/2012 3:00	3.7	-1.5	69	33	25.0	99.30	Mc Clo
...	...	...	...	...	...	...	...	
8705	12/28/2012 17:00	-8.6	-12.0	76	26	25.0	101.34	Ma C
8753	12/30/2012 17:00	-12.1	-15.8	74	28	25.0	101.26	Ma C
8755	12/30/2012 19:00	-13.4	-16.5	77	26	25.0	101.47	Ma C
8759	12/30/2012 23:00	-12.1	-15.1	78	28	25.0	101.52	Mc Clo
8760	12/31/2012 0:00	-11.1	-14.4	77	26	25.0	101.51	Clo

308 rows × 8 columns



10. Find the mean of all values against 'Weather Condition'

```
In [91]: data.groupby('Weather Condition' ).apply(lambda x: x)
```

C:\Users\hmmuz\AppData\Local\Temp\ipykernel_22952\1588354044.py:1: FutureWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

```
data.groupby('Weather Condition' ).apply(lambda x: x)
```

Out[91]:

		Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibili
Weather Condition							
Clear	67	1/3/2012 19:00	-16.9	-24.8	50	24	
	114	1/5/2012 18:00	-7.1	-14.4	56	11	
	115	1/5/2012 19:00	-9.2	-15.4	61	7	
	116	1/5/2012 20:00	-9.8	-15.7	62	9	
	117	1/5/2012 21:00	-9.0	-14.8	63	13	
...	...	...	...	...	...	...	...
Thunderstorms,Rain Showers	6188	9/14/2012 20:00	16.5	15.8	96	22	
Thunderstorms,Rain Showers,Fog	4323	6/29/2012 3:00	19.5	16.1	81	7	
	4761	7/17/2012 9:00	22.9	21.3	91	17	
	5108	7/31/2012 20:00	22.4	18.7	80	35	
Thunderstorms,Rain,Fog	4757	7/17/2012 5:00	20.6	18.6	88	19	

8784 rows × 8 columns



11. Find Minimum and Maximum value against the Weather Condition

```
In [93]: data.groupby('Weather Condition').min()
```

Out[93]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km
Weather Condition						
Clear	1/11/2012 1:00	-23.3	-28.5	20	0	11.3
Cloudy	1/1/2012 17:00	-21.4	-26.8	18	0	11.3
Drizzle	1/23/2012 21:00	1.1	-0.2	74	0	6.4
Drizzle,Fog	1/23/2012 20:00	0.0	-1.6	85	0	1.0
Drizzle,Ice Pellets,Fog	12/17/2012 9:00	0.4	-0.7	92	20	4.0
Drizzle,Snow	12/17/2012 15:00	0.9	0.1	92	9	9.7
Drizzle,Snow,Fog	12/18/2012 21:00	0.3	-0.1	92	7	2.4
Fog	1/1/2012 0:00	-16.0	-17.2	80	0	0.2
Freezing Drizzle	1/13/2012 10:00	-9.0	-12.2	78	6	4.8
Freezing Drizzle,Fog	1/1/2012 2:00	-6.4	-9.0	82	6	3.6
Freezing Drizzle,Haze	2/1/2012 11:00	-5.8	-8.3	81	9	2.0
Freezing Drizzle,Snow	1/13/2012 3:00	-8.3	-10.4	79	6	2.4
Freezing Fog	1/22/2012 6:00	-19.0	-22.9	71	0	0.2
Freezing Rain	1/13/2012 11:00	-6.5	-9.0	81	7	2.8
Freezing Rain,Fog	1/17/2012 23:00	-6.1	-8.7	82	7	2.8
Freezing Rain,Haze	2/1/2012 14:00	-4.9	-7.5	82	6	2.0
Freezing Rain,Ice Pellets,Fog	12/17/2012 3:00	-2.6	-3.7	92	28	8.0
Freezing Rain,Snow Grains	1/13/2012 9:00	-5.0	-7.3	84	32	4.8

Weather Condition	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km
Haze	1/22/2012 12:00	-11.5	-16.0	68	0	4.8
Mainly Clear	1/10/2012 11:00	-22.8	-28.0	20	0	12.9
Moderate Rain,Fog	12/10/2012 8:00	1.7	0.8	94	17	6.4
Moderate Snow	1/12/2012 15:00	-6.3	-7.6	83	26	0.6
Moderate Snow,Blowing Snow	12/27/2012 10:00	-5.5	-6.6	92	39	0.6
Mostly Cloudy	1/1/2012 16:00	-23.2	-28.5	18	0	11.3
Rain	1/1/2012 18:00	0.3	-5.7	40	0	4.0
Rain Showers	1/1/2012 22:00	1.6	-7.2	37	0	6.4
Rain Showers,Fog	10/20/2012 3:00	12.8	12.1	96	13	6.4
Rain Showers,Snow Showers	11/4/2012 8:00	2.1	-1.8	75	17	19.3
Rain,Fog	1/23/2012 18:00	0.0	-1.2	83	0	2.0
Rain,Haze	3/13/2012 7:00	4.0	1.0	81	7	4.0
Rain,Ice Pellets	12/18/2012 5:00	0.6	-0.6	92	24	9.7
Rain,Snow	1/10/2012 5:00	0.6	-1.7	81	13	2.4
Rain,Snow Grains	12/21/2012 0:00	1.9	-2.1	75	26	25.0
Rain,Snow,Fog	12/8/2012 21:00	0.8	0.3	96	9	6.4
Rain,Snow,Ice Pellets	12/21/2012 1:00	0.9	-0.7	88	17	4.8
Snow	1/10/2012 1:00	-16.7	-24.6	41	0	1.0

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km
Weather Condition						
Snow Pellets	11/24/2012 15:00	0.7	-6.4	59	35	2.4
Snow Showers	1/12/2012 7:00	-13.3	-19.3	52	0	2.4
Snow Showers,Fog	12/26/2012 9:00	-11.3	-12.7	89	7	4.0
Snow,Blowing Snow	1/13/2012 21:00	-12.0	-16.2	70	24	0.6
Snow,Fog	12/16/2012 15:00	-10.1	-12.0	77	4	1.2
Snow,Haze	2/1/2012 17:00	-4.3	-7.2	80	0	4.0
Snow,Ice Pellets	12/10/2012 3:00	-4.3	-5.9	76	19	2.8
Thunderstorms	7/16/2012 1:00	21.6	19.4	67	0	24.1
Thunderstorms,Heavy Rain Showers	5/29/2012 6:00	10.9	9.0	88	9	2.4
Thunderstorms,Moderate Rain Showers,Fog	7/17/2012 6:00	19.6	18.5	93	15	3.2
Thunderstorms,Rain	5/25/2012 20:00	19.4	18.2	83	4	16.1
Thunderstorms,Rain Showers	5/29/2012 16:00	11.0	7.0	68	7	6.4
Thunderstorms,Rain Showers,Fog	6/29/2012 3:00	19.5	16.1	80	7	9.7
Thunderstorms,Rain,Fog	7/17/2012 5:00	20.6	18.6	88	19	4.8

```
In [94]: data.groupby('Weather Condition').max()
```

Out[94]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km
Weather Condition						
Clear	9/9/2012 5:00	32.8	20.4	99	33	48.3
Cloudy	9/9/2012 23:00	30.5	22.6	99	54	48.3
Drizzle	9/30/2012 3:00	18.8	17.7	96	30	25.0
Drizzle,Fog	9/30/2012 2:00	19.9	19.1	100	28	9.7
Drizzle,Ice Pellets,Fog	12/17/2012 9:00	0.4	-0.7	92	20	4.0
Drizzle,Snow	12/19/2012 18:00	1.2	0.2	95	19	11.3
Drizzle,Snow,Fog	12/22/2012 3:00	1.1	0.6	98	32	9.7
Fog	9/22/2012 0:00	20.8	19.6	100	22	9.7
Freezing Drizzle	2/1/2012 5:00	-2.3	-3.3	93	26	12.9
Freezing Drizzle,Fog	12/10/2012 5:00	-0.3	-2.3	94	33	8.0
Freezing Drizzle,Haze	2/1/2012 13:00	-5.0	-7.7	83	11	4.0
Freezing Drizzle,Snow	3/2/2012 12:00	-3.3	-4.6	94	24	12.9
Freezing Fog	3/17/2012 6:00	-0.1	-0.3	99	9	0.8
Freezing Rain	2/1/2012 7:00	0.3	-1.7	92	28	16.1
Freezing Rain,Fog	12/17/2012 1:00	0.1	-0.9	93	26	9.7
Freezing Rain,Haze	2/1/2012 15:00	-4.9	-7.4	83	9	2.8
Freezing Rain,Ice Pellets,Fog	12/17/2012 3:00	-2.6	-3.7	92	28	8.0
Freezing Rain,Snow Grains	1/13/2012 9:00	-5.0	-7.3	84	32	4.8

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km
Weather Condition						
Haze	3/13/2012 23:00	14.1	11.1	86	17	9.7
Mainly Clear	9/9/2012 9:00	33.0	21.2	99	63	48.3
Moderate Rain,Fog	12/10/2012 8:00	1.7	0.8	94	17	6.4
Moderate Snow	12/27/2012 9:00	-4.9	-6.7	93	39	0.8
Moderate Snow,Blowing Snow	12/27/2012 12:00	-5.4	-6.4	93	41	0.6
Mostly Cloudy	9/9/2012 2:00	32.4	24.4	100	83	48.3
Rain	9/5/2012 2:00	22.8	20.4	99	52	48.3
Rain Showers	9/8/2012 16:00	26.4	23.0	97	41	48.3
Rain Showers,Fog	10/20/2012 3:00	12.8	12.1	96	13	6.4
Rain Showers,Snow Showers	12/5/2012 10:00	2.2	-1.2	78	28	24.1
Rain,Fog	9/30/2012 23:00	21.7	19.5	100	46	9.7
Rain,Haze	3/13/2012 9:00	5.5	2.9	86	17	9.7
Rain,Ice Pellets	12/18/2012 5:00	0.6	-0.6	92	24	9.7
Rain,Snow	4/23/2012 3:00	1.7	0.5	94	52	25.0
Rain,Snow Grains	12/21/2012 0:00	1.9	-2.1	75	26	25.0
Rain,Snow,Fog	12/8/2012 21:00	0.8	0.3	96	9	6.4
Rain,Snow,Ice Pellets	12/21/2012 5:00	1.3	0.1	94	28	6.4
Snow	4/27/2012 9:00	3.7	0.3	96	57	25.0

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km
Weather Condition						
Snow Pellets	11/24/2012 15:00	0.7	-6.4	59	35	2.4
Snow Showers	3/4/2012 21:00	2.9	-0.7	94	37	48.3
Snow Showers,Fog	12/29/2012 13:00	-10.0	-11.1	92	22	9.7
Snow,Blowing Snow	2/25/2012 9:00	-1.4	-2.9	91	48	9.7
Snow,Fog	3/14/2012 19:00	1.1	0.8	99	35	9.7
Snow,Haze	2/1/2012 21:00	-3.6	-6.4	81	15	6.4
Snow,Ice Pellets	3/3/2012 4:00	0.8	-1.7	92	33	11.3
Thunderstorms	7/4/2012 16:00	26.7	20.1	87	15	25.0
Thunderstorms,Heavy Rain Showers	5/29/2012 6:00	10.9	9.0	88	9	2.4
Thunderstorms,Moderate Rain Showers,Fog	7/17/2012 6:00	19.6	18.5	93	15	3.2
Thunderstorms,Rain	7/23/2012 18:00	21.3	19.1	93	30	24.1
Thunderstorms,Rain Showers	9/8/2012 4:00	25.5	23.1	98	32	25.0
Thunderstorms,Rain Showers,Fog	7/31/2012 20:00	22.9	21.3	91	35	9.7
Thunderstorms,Rain,Fog	7/17/2012 5:00	20.6	18.6	88	19	4.8

12. Find all where Weather is 'Clear' or Visibility is more than 40

```
In [99]: data[(data['Weather Condition'] == 'Clear') | (data['Visibility_km'] > 40)]
```


Out[99]:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weat Condit
67	1/3/2012 19:00	-16.9	-24.8	50	24	25.0	101.74	C
106	1/5/2012 10:00	-6.0	-10.0	73	17	48.3	100.45	Ma C
107	1/5/2012 11:00	-5.6	-10.2	70	22	48.3	100.41	Ma C
108	1/5/2012 12:00	-4.7	-9.6	69	20	48.3	100.38	Ma C
109	1/5/2012 13:00	-4.4	-9.7	66	26	48.3	100.40	Ma C
...	...	...	...	...	...	...	...	
8749	12/30/2012 13:00	-12.4	-16.2	73	37	48.3	100.92	Mc Clo
8750	12/30/2012 14:00	-11.8	-16.1	70	37	48.3	100.96	Ma C
8751	12/30/2012 15:00	-11.3	-15.6	70	32	48.3	101.05	Ma C
8752	12/30/2012 16:00	-11.4	-15.5	72	26	48.3	101.15	Ma C
8756	12/30/2012 20:00	-13.8	-16.5	80	24	25.0	101.52	C

3027 rows × 8 columns



13. Find all where Weather = Clear and Relative Humidity > 50
or
Visibility > 40

In [101...

data[(data ['Weather Condition'] == 'Clear') & (data['Rel Hum_%'] > 50) | (data ['V

Out[101...

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weat Condit
106	1/5/2012 10:00	-6.0	-10.0	73	17	48.3	100.45	Ma C
107	1/5/2012 11:00	-5.6	-10.2	70	22	48.3	100.41	Ma C
108	1/5/2012 12:00	-4.7	-9.6	69	20	48.3	100.38	Ma C
109	1/5/2012 13:00	-4.4	-9.7	66	26	48.3	100.40	Ma C
110	1/5/2012 14:00	-5.1	-10.7	65	22	48.3	100.46	Ma C
...	...	...	...	...	...	...	...	
8749	12/30/2012 13:00	-12.4	-16.2	73	37	48.3	100.92	Mc Clo
8750	12/30/2012 14:00	-11.8	-16.1	70	37	48.3	100.96	Ma C
8751	12/30/2012 15:00	-11.3	-15.6	70	32	48.3	101.05	Ma C
8752	12/30/2012 16:00	-11.4	-15.5	72	26	48.3	101.15	Ma C
8756	12/30/2012 20:00	-13.8	-16.5	80	24	25.0	101.52	C

2921 rows × 8 columns



In []: